



DQ Dairy-Free

A CONJOINT STUDY ON NON-DAIRY ICE CREAM PREFERENCES

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Executive Summary (Anirudh)

Consumer preferences across the quick-service restaurant landscape are shifting materially toward plant-based and non-dairy alternatives, driven by lactose intolerance, veganism, environmental consciousness, and a broader flexitarian trend. Competitors have already moved aggressively into this space, and continued inaction risks ceding a growing and loyal segment, particularly Millennials and Gen Z, to more progressive rivals. Beyond dietary preference, food choices increasingly function as value signaling, with consumers aligning purchases with causes such as animal welfare and climate advocacy. This social dimension adds both urgency and complexity to Dairy Queen's positioning decision.

To quantify how consumers weigh product attributes when evaluating non-dairy frozen dessert options, a conjoint analysis was conducted across key variables including price point, non-dairy base ingredient, flavor profile, availability format, and values-based messaging. The analysis surfaces clear consumer preference hierarchies and identifies the attribute combinations most likely to drive trial and purchase among the target segment.

The findings reveal actionable sweet spots that allow Dairy Queen to introduce non-dairy offerings without compromising its value positioning. Both drive-thru and walk-in formats generate similarly high consumer utility, reducing the need for costly channel investment. On price, the analysis identifies a viable range that balances the premium ingredient costs inherent to non-dairy production with the accessibility expectations central to the brand promise.

The recommended strategy positions Dairy Queen to enter the non-dairy segment in a way that is economically viable, brand-consistent, and credible to the consumers it most needs to reach, grounded in measured preference data rather than assumption.

Background Summary (Elizabeth)

Dairy Queen (DQ) is one of the most recognized quick-service restaurant brands in the world, operating over 7,000 locations across more than 20 countries under the ownership of Berkshire Hathaway. Founded in 1940, the brand has built its identity around soft-serve ice cream, Blizzard treats, and accessible, indulgent dessert offerings. Despite this legacy, the quick service restaurant landscape is undergoing a fundamental shift driven by changing consumer preferences, with the

plant-based and non-dairy food market experiencing rapid growth across all generational segments. Consumers are increasingly seeking alternatives to traditional dairy products due to lactose intolerance, veganism, environmental consciousness, and a broader flexitarian trend and competitors such as McDonald's, Burger King, and specialty dessert chains have already moved aggressively into this space. If Dairy Queen fails to respond, it risks ceding a growing and loyal customer segment, particularly Millennials and Gen Z, to more progressive rivals.

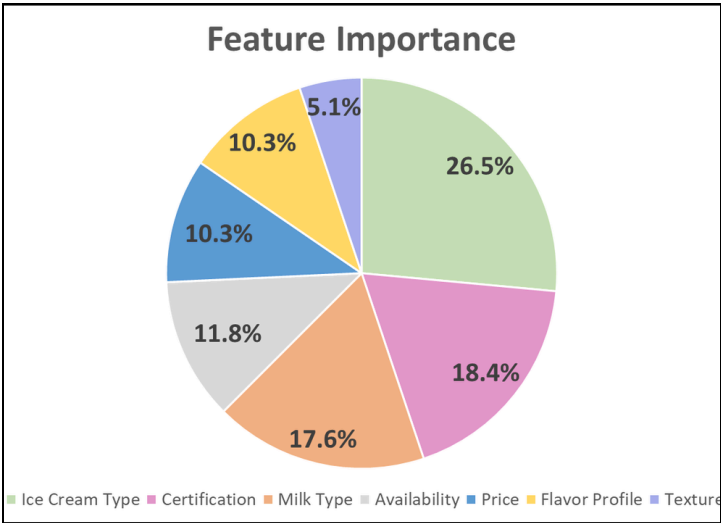
Beyond dietary trends, a social dimension has emerged as a significant driver of non-dairy adoption. Consumers increasingly use food choices as a form of value signaling, aligning purchases with causes such as animal welfare, climate change mitigation, and personal health advocacy. For Dairy Queen, this social element represents both a challenge and an opportunity: the brand must determine how to authentically communicate non-dairy offerings without alienating its core customer base. Compounding this is a pricing tension central to DQ's brand promise. Non-dairy alternatives historically carry a premium due to ingredient costs, yet Dairy Queen's identity is built on value and accessibility. The company must therefore determine not just whether to offer non-dairy products, but how to price and position them in a way that is economically viable while remaining genuinely compelling to target consumers.

Dairy Queen recognizes that consumers place varying levels of importance on product attributes such as price point, ingredient transparency, taste parity, and social or environmental messaging when evaluating non-dairy frozen dessert options. To quantify these preferences, a conjoint analysis was conducted presenting respondents with product profiles varying across key attributes including price, non-dairy base ingredient, and values-based messaging. The findings will enable Dairy Queen to make data-driven decisions about how to introduce, price, and position non-dairy options in a way that maximizes consumer appeal, supports an affordable positioning, and leverages the social motivations that increasingly drive purchase behavior.

Findings (Haydar)

Overall Feature Importance

The "Feature Importance" pie chart illustrates how the seven product attributes studied in the conjoint analysis were broken down into relative amounts of decision weight. It was found that ice cream type accounted for the greatest amount of decision weight with 26.5 percent, certification with 18.4 percent and milk type with 17.6 percent. Together, the three most important factors accounted for 62.5% of all feature importance; this indicates that consumers' evaluations of non-dairy frozen desserts are largely dependent upon the format of the product, the credentials associated with the product and the base ingredient used in the product.



The next most important factor was availability, accounting for an additional 11.8%, thus making the cumulative importance of the top four features 74.3%. This indicates that the remaining three factors, although not insignificant, represent secondary influences on consumers' preferences.

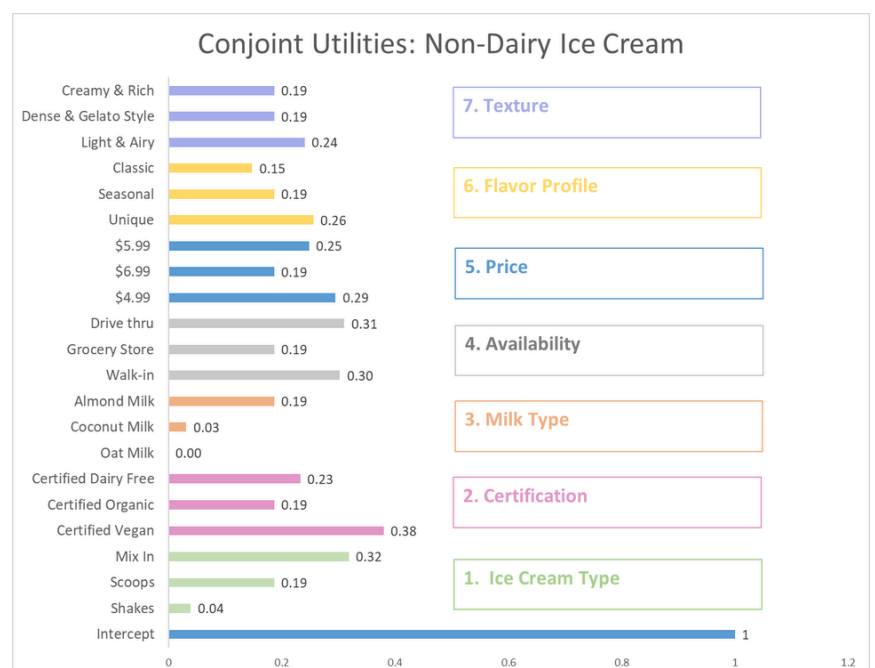
Availability (11.8%) and price (10.3%), along with flavor profile (10.3%) represented the middle tier of factors. In this case, both price and flavor profile represented equal weights of influence; however, based upon this finding alone it is also worthy to note that consumers in this sample did not distinguish between cost-related factors and flavor-related factors to any greater extent than they distinguished between either of those two factors and where the product could be purchased. Texture (5.1%), represented less than half of the importance placed on ice cream type, and therefore indicated that taste, while contributing to the total experience of the product, was not a primary factor influencing consumer purchasing behavior within this category.

When taken collectively, the data illustrated by the pie chart clearly demonstrates a hierarchical structure in which the structural/credential-based attributes have greater importance than the sensory-based attributes. Therefore, Dairy Queen's product configuration choices should focus their efforts first on ice cream type, certification, and milk type as small changes in levels selected will result in large changes in overall consumer preference.

Conjoint Utility Value

The Conjoint Utility Bar Graph measures the partial utility of every level of the 7 attributes. All utilities will be compared to a Model Intercept of 1.00. When reviewing the attributes individually, we see where consumers' preference concentrate, and also identify some significant results worthy of additional scrutiny.

Ice Cream Type is the most important attribute on the chart. As such, when considering the highest utility generated by each attribute, Mix-ins show the highest utility at 0.32, followed closely



by scoops at 0.19 and then shakes at 0.04. Additionally, the 0.28 point difference between mix-ins and shakes is the largest differentials on the chart among attribute levels, and this is consistent with Mix-in's greater influence in the importance analysis. On the other hand, Shakes has a very low utility rating. A format that is at the heart of several competitor menu formats provides little to no incremental utility above the base line; therefore, it appears that Shakes may not be attractive as a leading non-dairy format for this segment.

Milk-type has the most unexpected outcome on the chart. Almond milk has the highest utility (.19), while coconut milk has .03 and oat milk has 0.00. While oat milk was surprising given the cultural momentum in recent years across coffee, yogurt and packaged ice cream in terms of oat-based alternatives, there is no evidence that this same trend exists in terms of non-dairy frozen desserts. Likewise, coconut milk performed much better than oat milk. Given the small sample size, this finding needs to be viewed cautiously; however based on the data available, only almond milk appears to provide meaningful movement in this regard.

As expected, Price behaved inversely. Utility decreased as prices increased. The highest rated price point \$4.99 produced the highest utility .29, while \$5.99 was second at .25 and \$6.99 was last at .19. There is a 0.04 point differential between \$4.99 and \$5.99; thus while a dollar increase from the least expensive tier resulted in only a moderate decrease in preference.

Flavor Profile shows that the way that people will respond to new or unique items on Dairy queen's menu is stronger than to products which have seasonal flavors and even weaker when they are classics. Therefore, it demonstrates the strength of developing a positioning strategy using innovation rather than creating a long-term formulation. However, the lower utility scores within this attribute are consistent with consumers weighing the flavor profile component less heavily than other components in the hierarchy of their importance.

There were several discrepancies identified within the data which indicate that there may be some follow-up required. These include: Oat Milk has no reported utility, although oat milk has had tremendous momentum within its category; Shakes have been shown to have extremely low utility as a format; There was an equal utility produced by Dense & Gelato Style and Creamy & Rich textures; Drive Thru and Walk-In Availability show almost equal utilities. Each of these outcomes can have significant effects on the configuration of each product offering and would likely require validation via future research if larger sample sizes are utilized.

Suggested Strategies (Natalia)

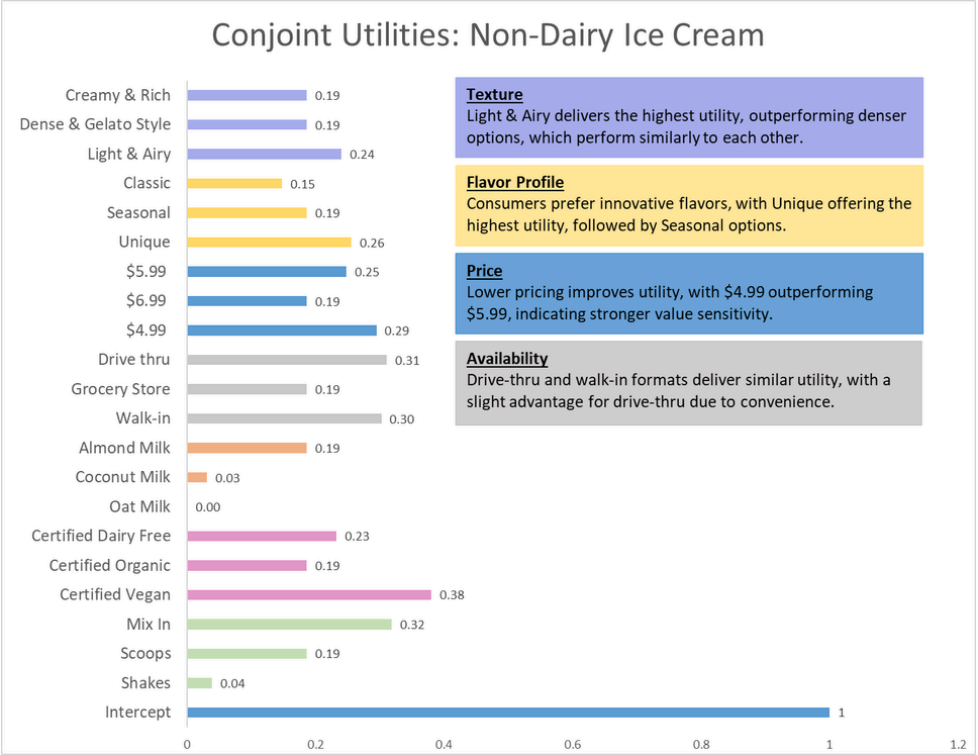
Dairy Queen's recommended strategy is to reinforce its leadership in a mature market by prioritizing the highest-impact attributes while making targeted trade-offs to balance profitability and brand experience. The conjoint results clearly indicate that consumer preference is driven most by certification, texture, flavor differentiation, and accessibility, so the non-dairy line extension should anchor on a light & airy texture, certified vegan positioning, and unique flavor offerings to maximize utility and maintain competitive advantage. While a \$4.99 price point yields the highest utility, shifting to \$5.99 enables stronger margins and reinforces a more premium, experience-driven product, with only a marginal loss in preference. In terms of distribution, although both walk-in and drive-thru formats fall within the same "sweet spot," prioritizing walk-in locations allows Dairy Queen to differentiate through a more social, in-store experience, rather than competing solely on convenience. This is especially important given that competitors like Baskin-Robbins and Andy's Frozen Custard already perform strongly in drive-thru accessibility. By aligning with key sweet spots while making intentional trade-offs, Dairy Queen can sustain its leading TPU and market share position, while building a more distinctive and profitable long-term strategy against competitors such as Jeni's and Ben & Jerry's.

Sweet Spots

The conjoint utility analysis reveals "sweet spots" where different feature combinations deliver similar overall utility (differences ≤ 0.1), indicating multiple viable product configurations rather than a single dominant option. Across attributes, the key sweet spots are:

Availability: Both drive-thru and walk-in options generate similarly high utilities. This indicates a sweet spot where offering either format effectively satisfies consumer demand for convenience and access, with minimal difference in perceived value between the two channels.

Price: While \$5.99 per unit can perform reasonably well, pricing at \$4.99 per unit meaningfully increases utility. Higher price tiers show a clear drop-off in consumer preference, reinforcing the importance of a strong value-led positioning strategy.



Flavor Profile: Unique and Seasonal flavor profiles form the primary sweet spot, with both delivering comparable utility within the ≤ 0.1 threshold. This indicates that consumers respond similarly to both innovation-led and limited-time offerings, allowing flexibility in execution. Classic flavors, while still viable, fall into a lower tier and are less central to optimal configurations.

Texture: Creamy & Rich vs Dense & Gelato style textures offer the same utility, forming a clear sweet spot where either texture direction can be pursued without meaningful differences in consumer preference. However, Light & Airy outperforms both alternatives, making it the preferred option for maximizing overall utility. Aligning product formulation toward this lighter texture can further strengthen market positioning while still retaining flexibility within the secondary texture sweet spot.

Trade-offs of Feature Types

While the conjoint results identify clear preferred levels across attributes, implementing the highest-utility level for every feature may be operationally or financially inefficient. As such, strategic trade-offs are required to maximize overall utility while maintaining cost efficiency and feasibility.

For Dairy Queen, the optimal strategy prioritizes reinforcing brand equity. The non-dairy line extension will retain Light & Airy texture as a core non-negotiable, as it best aligns with both the highest-performing texture attribute and Dairy Queen’s established brand identity. To further enhance perceived value and experience, the strategy will emphasize Unique flavor profiles, reinforcing innovation and variety as key demand drivers.

To support this positioning, price will be adjusted from \$4.99 to \$5.99 per unit, accepting a moderate reduction in price utility in exchange for stronger profit margin and premium signaling that better reflects an elevated product experience.

In addition, while drive-thru and walk-in channels deliver similar utility, the strategy will prioritize walk-in locations to enhance the social and experiential aspect of Dairy Queen stores, shifting emphasis from convenience-only

access to in-store engagement.

TPU & Market Share

In this mature market (intercept = 1), where all brands begin on relatively equal footing, differences in Total Preference Utility (TPU) translate directly into shifts in market share. Dairy Queen leads with the highest TPU (1.76), which corresponds to the largest market share at 22.77%, driven by strong alignment with high-utility attributes such as certified vegan status, walk-in availability, and a light & airy, unique product profile. Jeni's follows closely (TPU 1.74) with a 21.98% share, as its similar strengths are slightly offset by a higher price point. Baskin-Robbins (TPU 1.54) captures 19.53% of the market by leveraging value pricing and drive-thru convenience, though its more traditional positioning limits differentiation. Andy's Frozen Custard (TPU 1.47) holds 18.65% share, supported by strong mix-in appeal and accessibility, but constrained by lower-utility milk options. Finally, Ben & Jerry's has the lowest TPU (1.35) and the smallest share at 17.08%, as weaker attributes like grocery store availability and coconut milk reduce its overall appeal despite strong flavor innovation. Overall, in a mature market, even modest differences in utility lead to meaningful differences in market share, reinforcing the importance of optimizing the most valued product attributes.

Dairy Queen	
Utilities	Optimal TPU
Light & Airy	0.24
Unique	0.25
\$5.99	0.24
Walk In	0.3
Almond	0.18
Certified Vegan	0.37
Scoops	0.18
	1.76

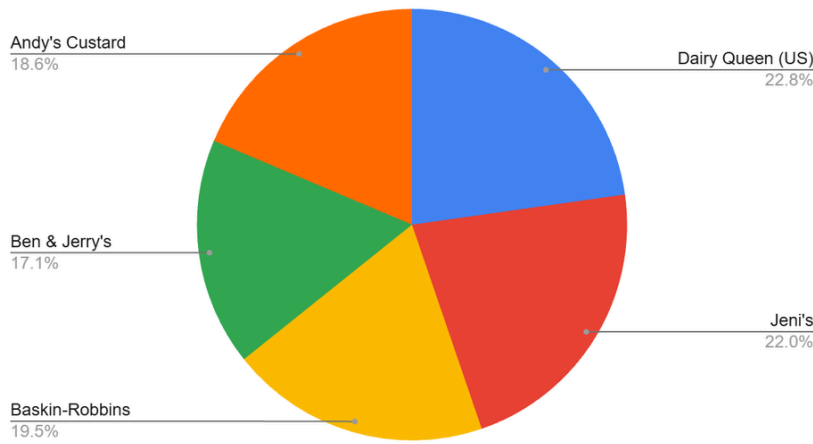
Jeni's	
Utilities	TPU
Light & Airy	0.24
Unique	0.26
\$6.99	0.19
Walk-In	0.30
Almond Milk	0.19
Cert. Vegan	0.38
Scoops	0.19
	1.74

Baskin-Robbins	
Utilities	Optimal TPU
Creamy & Rich	0.19
Classic	0.15
\$4.99	0.29
Drive Thru	0.31
Almond Milk	0.19
Cert. Dairy Free	0.23
Scoops	0.19
	1.54

Ben & Jerry's	
Utilities	Optimal TPU
Creamy & Rich	0.19
Unique	0.26
\$6.99	0.19
Grocery Store	0.19
Coconut Milk	0.03
Cert. Organic	0.19
Mix-Ins	0.32
	1.35

Andy's Custard	
Utilities	Optimal TPU
Dense & Gelato	0.19
Classic	0.15
\$5.99	0.25
Drive Thru	0.31
Coconut Milk	0.03
Cert. Dairy Free	0.23
Mix-Ins	0.32
	1.47

Market Share



Conjoint Analysis (Naveen)

Tool Overview

Description

For this study, conjoint analysis was used to quantify how consumers trade off key product attributes when evaluating non-dairy ice cream options. Rather than directly asking consumers what they prefer, the method infers preference by analyzing how respondents evaluate different product combinations.

This approach allows Dairy Queen to identify which attributes drive purchase intent, measure their relative importance, and determine the optimal product configuration based on real trade-offs.

Mechanics

- **STEP 1: Segment and Problem** - The target segment consists of Millennials and Gen Z consumers who are health-conscious, lactose intolerant, vegan, or reducing dairy consumption. The core problem is to determine the optimal combination of attributes for a non-dairy ice cream product that maximizes purchase intent.
- **STEP 2: Features and Level Selection** - Seven features with three levels each were selected through a structured team exercise informed by industry trends and course concepts. These features capture functional, experiential, and values-based aspects of consumer decision-making.
- **STEP 3: Choose Orthogonal Design** - An L18 orthogonal design was used to reduce the total possible combinations ($3^7 = 2,187$) into 18 manageable product profiles while preserving statistical independence between attributes.

Product ID	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10	x11	x12	x13	x14
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1	0	1	0	1	0	1	0	1	0	1	0	0	0
3	0	1	0	1	1	0	1	0	1	0	1	0	1	0
4	0	0	0	0	0	1	0	0	1	1	0	0	1	0
5	1	0	1	0	0	0	1	0	0	0	1	0	0	0
6	0	1	0	0	1	0	0	1	0	0	0	1	0	0
7	0	0	1	0	0	0	0	0	1	0	1	1	0	1
8	1	0	0	1	1	1	0	0	0	0	0	0	1	0
9	0	1	0	0	0	0	1	1	0	1	0	0	0	1
10	0	0	0	0	1	0	1	0	0	1	0	1	0	0
11	1	0	0	0	0	0	0	1	0	0	1	0	1	0
12	0	1	1	0	0	1	0	0	1	0	0	0	0	1
13	0	0	1	0	0	0	1	1	0	0	0	0	1	0
14	1	0	0	0	1	0	0	0	1	1	0	0	0	1
15	0	1	0	0	0	1	0	0	0	0	1	1	0	0
16	0	0	0	0	1	1	0	1	0	0	1	0	0	0
17	1	0	0	0	0	0	1	0	1	0	0	1	0	0
18	0	1	1	0	0	0	0	0	0	1	0	0	1	0

- **STEP 4: Create Codebook** - A dummy variable coding scheme was created, assigning binary variables to each feature level while maintaining a reference level. This enabled regression-based estimation of utility values.

Code Sheet		
Price	x1=0,1	0= NOT 4.99, 1= 4.99
	x2=0,1	0= NOT 5.99, 1= 5.99
Availability	x3=0,1	0= NOT WALK INS, 1= WALK-IN
	x4=0,1	0= NOT DRIVE THURS, 1= DRIVE THURS
Non-Dairy Milk type	x5=0,1	0= NOT OAT MILK, 1= OAT MILK
	x6=0,1	0= NOT COCONUT MILK, 1= COCONUT MILK
Types of ice cream	x7=0,1	0= NOT MIX IN, 1= MIX IN
	x8=0,1	0= NOT SHAKES, 1= SHAKES
Flavor Profile	x9=0,1	0= NOT CLASSIC, 1= CLASSIC
	x10=0,1	0= NOT UNIQUE, 1= UNIQUE
Certification	x11 = 0,1	0= NOT CERTIFIED VEGAN, 1= CERTIFIED VEGAN
	x12 = 0,1	0= NOT CERTIFIED DAIRY FREE, 1= CERTIFIED DAIRY FREE
Texture/Consistency	x13 = 0,1	0= NOT LIGHTS & AIRY, 1= LIGHT & AIRY
	x14 = 0,1	0= NOT CREAMY & RICH, 1= CREAMY & RICH

- **STEP 5: Design Survey** - Respondents evaluated all 18 product profiles using a 7-point purchase likelihood scale (1 = would not purchase, 7 = would strongly consider). The structure ensured consistent comparison across attribute combinations.

DAIRY QUEEN — NON-DAIRY ICE CREAM CONJOINT ANALYSIS					
Survey Profiles — L18 Orthogonal Design Rate each product 1-7: 1=Would definitely NOT purchase → 7=Would definitely consider purchasing					
SAMPLE		Rating Scale:			
		1. Would definitely NOT purchase 2. Would most likely NOT purchase 3. Would probably NOT purchase 4. Indifferent towards purchasing 5. Would probably consider purchasing 6. Would most likely consider purchasing 7. Would definitely consider purchasing			
Product 1		Product 2		Product 3	
Price	\$6.99	Price	\$4.99	Price	\$5.99
Availability	Grocery Store	Availability	Walk-in	Availability	Drive-thru
Milk Type	Almond Milk	Milk Type	Oat Milk	Milk Type	Coconut Milk
Type of Ice Cream	Scoop	Type of Ice Cream	Mix-in	Type of Ice Cream	Shakes
Flavor Profile	Seasonal	Flavor Profile	Classic	Flavor Profile	Unique
Certification	Organic	Certification	Vegan	Certification	Dairy Free
Texture/Consistency	Light & Airy	Texture/Consistency	Creamy & Rich	Texture/Consistency	Creamy & Rich
Product 7		Product 8		Product 9	
Price	\$6.99	Price	\$4.99	Price	\$5.99
Availability	Walk-in	Availability	Drive-thru	Availability	Grocery Store
Milk Type	Almond Milk	Milk Type	Oat Milk	Milk Type	Coconut Milk
Type of Ice Cream	Shakes	Type of Ice Cream	Scoop	Type of Ice Cream	Mix-in
Flavor Profile	Unique	Flavor Profile	Seasonal	Flavor Profile	Classic
Certification	Vegan	Certification	Dairy Free	Certification	Organic
Texture/Consistency	Light & Airy	Texture/Consistency	Light & Airy	Texture/Consistency	Light & Airy
Product 13		Product 14		Product 15	
Price	\$6.99	Price	\$4.99	Price	\$5.99
Availability	Walk-in	Availability	Drive-thru	Availability	Grocery Store
Milk Type	Coconut Milk	Milk Type	Almond Milk	Milk Type	Oat Milk
Type of Ice Cream	Mix-in	Type of Ice Cream	Shakes	Type of Ice Cream	Scoop
Flavor Profile	Seasonal	Flavor Profile	Classic	Flavor Profile	Unique
Certification	Dairy Free	Certification	Organic	Certification	Vegan
Texture/Consistency	Creamy & Rich	Texture/Consistency	Creamy & Rich	Texture/Consistency	Creamy & Rich
Product 16		Product 17		Product 18	
Price	\$6.99	Price	\$4.99	Price	\$5.99
Availability	Drive-thru	Availability	Grocery Store	Availability	Walk-in
Milk Type	Oat Milk	Milk Type	Coconut Milk	Milk Type	Almond Milk
Type of Ice Cream	Mix-in	Type of Ice Cream	Shakes	Type of Ice Cream	Scoop
Flavor Profile	Unique	Flavor Profile	Seasonal	Flavor Profile	Classic
Certification	Organic	Certification	Vegan	Certification	Dairy Free
Texture/Consistency	Creamy & Rich	Texture/Consistency	Creamy & Rich	Texture/Consistency	Creamy & Rich

- **STEP 6: Collect Data** - The survey was completed by 5 respondents, generating a total of 90 observations (5 respondents × 18 profiles). While the sample size is limited, it is sufficient for directional insights in a controlled academic setting.

- **STEP 7: Analyze Data** - Ordinary least squares regression was used to estimate utility values for each feature level. These utilities were rescaled to a 0–1 range for comparability. Feature importance was calculated based on the range of utilities within each attribute.

*(Significance not less than 0.05 since the sample size is small and our preferences may cancel each other out as a result)

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.413541							
R Square	0.171016							
Adjusted R Square	0.016272	10%-30% acceptable, >30% awesome						
Standard Error	1.920879							
Observations	90							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	14	57.08889	4.077778	1.105155	0.367591	Not < 0.05		
Residual	75	276.7333	3.689778					
Total	89	333.8222						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	3.5	0.784196	4.463171	2.80E-05	1.937802	5.062198	1.937802	5.062198
X Variable 1	0.466667	0.495969	0.940919	0.349767	-0.52135	1.454687	-0.52135	1.454687
X Variable 2	0.266667	0.495969	0.537668	0.592398	-0.72135	1.254687	-0.72135	1.254687
X Variable 3	0.5	0.495969	1.008128	0.316635	-0.48802	1.488021	-0.48802	1.488021
X Variable 4	0.533333	0.495969	1.075336	0.285671	-0.45469	1.521354	-0.45469	1.521354
X Variable 5	-0.8	0.495969	-1.613	0.110944	-1.78802	0.188021	-1.78802	0.188021
X Variable 6	-0.66667	0.495969	-1.34417	0.182945	-1.65469	0.321354	-1.65469	0.321354
X Variable 7	0.566667	0.495969	1.142545	0.256862	-0.42135	1.554687	-0.42135	1.554687
X Variable 8	-0.63333	0.495969	-1.27696	0.205555	-1.62135	0.354687	-1.62135	0.354687
X Variable 9	-0.16667	0.495969	-0.33604	0.737777	-1.15469	0.821354	-1.15469	0.821354
X Variable 10	0.3	0.495969	0.604877	0.547087	-0.68802	1.288021	-0.68802	1.288021
X Variable 11	0.833333	0.495969	1.680213	0.097077	-0.15469	1.821354	-0.15469	1.821354
X Variable 12	0.2	0.495969	0.403251	0.687911	-0.78802	1.188021	-0.78802	1.188021
X Variable 13	0.233333	0.495969	0.47046	0.639392	-0.75469	1.221354	-0.75469	1.221354
X Variable 14	-4.90E-16	0.495969	-9.90E-16	1	-0.98802	0.988021	-0.98802	0.988021

- **STEP 8: Interpret Results and Solve Problem** - The results identify the highest-value product configuration, highlight key trade-offs, and reveal “sweet spots” where multiple configurations deliver similar utility. These insights directly inform Dairy Queen’s product design, pricing strategy, and positioning in the non-dairy segment.